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Wearable Devices for Monitoring Work related Musculoskeletal and Gait Disorders

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***Abstract*— Wearable devices are gaining attention due to their capability to continuously monitor human health conditions and enable early detection of diseases. Numerous wearable devices have been developed for detecting physiological signals and monitoring of physical activities which strongly affect the quality of human life. The gait pattern of an individual strongly depends on their occupation, mood and age. Gait and posture monitoring is important for muscle assessment and motion, joint kinematics and physical examination. Such monitoring helps to prevent work related musculoskeletal disorders (WMSDs) and chronic disability in elderly people due to the gait disorders and related fall incidents. In this review, contemporary progress on the sensors developments implemented in wearable devices for monitoring WMSDs and gait disorders is summarized.**

Keywords— Wearable devices; Sensors; Flexible devices; Physiological parameters; Gait disorder; Musculoskeletal disorder

I. Introduction

In recent years, the healthcare sector has been hugely impacted by wearable electronics technology. Wearable devices are capable of the real time monitoring of multiple physiological parameters (e.g., respiration [1], pulse [2], temperature [3], pressure [4], humidity [5], ECG [6] and strain [7]) and human physical movements or physical activities (e.g., running, walking, jogging, standing and sitting) [8-10]. Wearables in combination with wireless communication technology offer remote monitoring. of diseases and serious health conditions that can be prevented or alleviated by early detection. Wearable technology has been identified as one of the key technologies by European Parliament Scientific & Technology Options Assessment Panel [11]. For accurate sensing and recording of reliable signals, the contact between the human body skin and the wearable devices is important. Numerous flexible wearable devices have been developed to monitor human physiological signals and physical activities.

In day-to-day life environment, physical activity monitoring is very important to understand and prevent the work related musculoskeletal disorders (WMSDs) and postural orientation. These conditions reduce the standard of living and result a huge number of population facing negative economic consequences, at individual, enterprise and societal levels [12]. A single or combination of work patterns such as continual repetitions of movements, prolonged awkward static/repetitive postures, concentrated force over small body regions (e.g. wrist, lower back or neck), long working hours, and work paces without sufficient recovery intervals are major risk factors for occurrence of WMSDs. For both office and physically demanding jobs, the prevalence of WMSDs is the leading cause of work disability. Similar to WMSDs, gait disorders and related fall incidents are major reasons of chronic disability in the elderly people [13, 14]. The structural or biomechanical abnormalities in human feet or lower limbs can be determined via gait analysis. A wearable fall detection device generally consists of accelerometer, gyroscopes, magnetic sensors, and a processing unit. An insole mounted wearable sensor system can be used in many applications including gait monitoring, posture and activity recognition, fall risk analysis, plantar pressure measurement, energy expenditure estimation and generation of biofeedback for assessing posture instability or gait disturbances [15-20].

Wearable devices have a huge potential for positive impact within the healthcare sector. This field is of increasing interest to the research community as evidenced by the growing research efforts and developments. Wearable devices are capable of non-invasive and cost effective daily health state monitoring and

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disease prevention. This review paper summarizes the recent developments of wearable devices for monitoring physiological signals and physical activities that strongly affect WMSDs and gait disorders.

II. SENSORS FOR WEARABLE WORK RELATED MUSCULOSKELETAL DISORDER (WMSD) DETECTION DEVICES

Many employees in all kinds of job fields affect work related musculoskeletal disorders (MSDs) in each year. MSDs include a number of health disorders and conditions. One main health problem is repetitive strain injuries (RSI), i.e., the discomfort in tendons, nerves, and muscles, by the overuse and repeated activities or movements for a long period in the absence of rest or poor posture. The RSI is also referred to work-related upper limb disorder. Lower limbs can also be affected by repetitive movements. These disorders are connected with particular occupations or job or activity, such as manual handling, working on assembly lines, computer usage and driving for prolonged hours. The resulting health problems from these activities range from minor pains to severe medical conditions. Physiological signals such as temperature, pulse rate, respiration rate, humidity, motion etc. are correlated with human physical activities. Therefore, work related health injuries can be prevented by monitoring these physiological signals.

Wearable and flexible sensor devices have been developed for detecting physiological parameters including heart rate [21], respiratory rate [22], blood pressure [23], skin conductance [24] and body temperature [25]. Very recently, Yi-FeiWang et al., developed a wearable device as shown in Fig 1 based on a fully printed temperature sensor with enhanced temperature sensitivity and humidity stability [26]. The electrodes of this temperature sensor were inkjet printed on Polyethlene Naphthalate (PEN) substrate using silver nanoparticle ink. Here, PEDOT:PSS with (3-glycidyloxypropyl) trimethoxysilane (GOPS) polymer was used as the sensing layer. The developed wearable temperature sensing device consisted of resistors, capacitors, an operational amplifier on a flexible circuit, and a wireless bluetooth low energy module. All these parts were connected using silver paste by screen-printing technique. A wearable leather-based respiration humidity sensor developed by Ruijie Xie et al. in 2019, is reported to be able to non-invasively detect human respiration depth, respiration rate, and respiration pattern (Fig. 2) [27]. Authors suggested this cost effective and biocompatible sensor for routine and at-home monitoring as this enables precise and continuous monitoring of respiration activity including apnea, deep breath, and normal

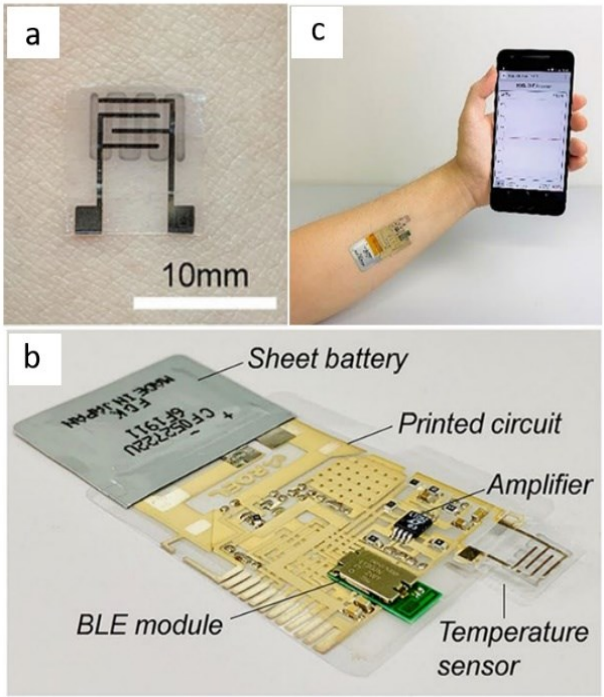

Fig. 1. *Wearable temperature sensing device (a) Image of the printed temperature sensor (b) Photograph of temperature sensing platform and (c) sensor platform worn on the human body skin [26]. Copyright © 2020, Springer Nature.*

breath. Metal oxide gas sensors are cable to measure gas in parts per billion range [28, 29]. Therefore, they provide a highly sensitive method for measuring breath rate [30]. Screen printing technique enables metal oxide gas sensors print these sensors on flexible substrates and hence can be used in the development of wearable devices.

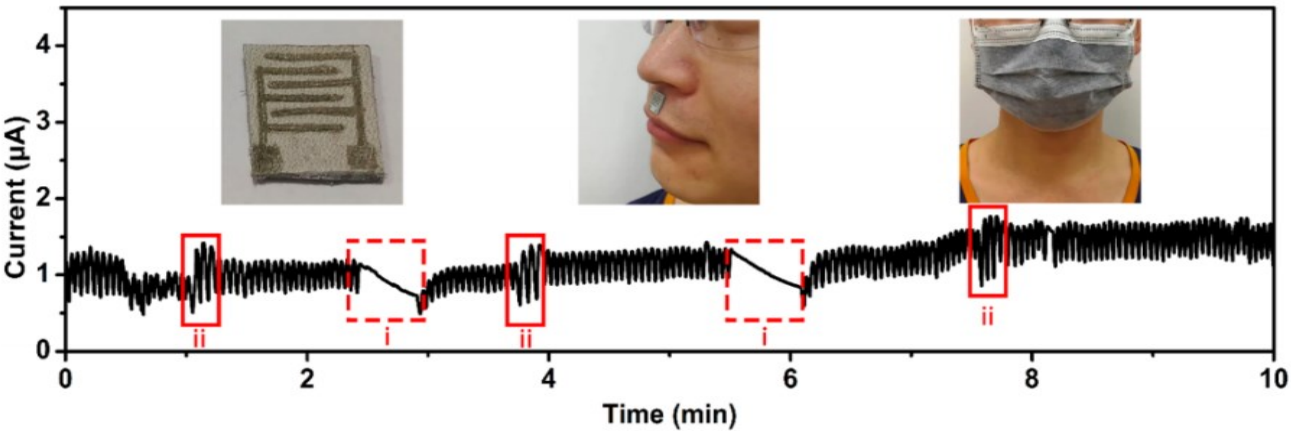

Fig. 2. *Respiration activity monitoring by attaching sensor under nose and a surgical mask [27]. Copyright © 2019, American Chemical Society.*

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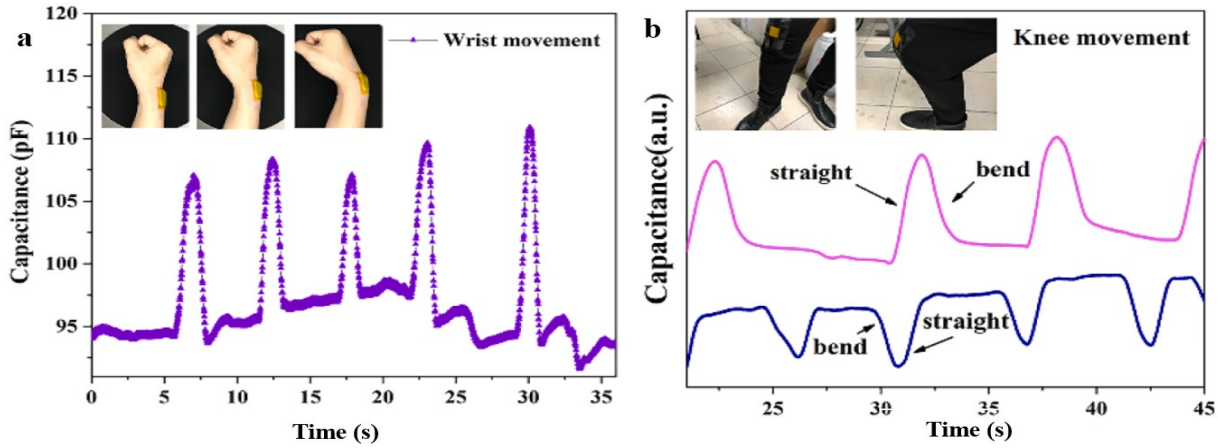

Fig. 3. *The human motion detection of the developed capacitive sensor by attaching (a) wrist and (b) knee with the capacitance variation as a function of time [31].*

A microporous pressure sensor based on capacitive operating mechanism exhibiting sensitivity 1.49 kPa−1 (< 1 kPa) and 0.13 kPa−1 (1–10 kPa) with a lower hysteresis (7.55%) has been fabricated for human motion monitoring (Fig. 3) [31]. The indicators used to evaluate pressure sensor performance were response time, durability and sensitivity. Human motion detection using this sensor was studied by attaching on the knee and wrist. This pressure sensor was based on Pd2+/PAN mixed solution and it exhibited rapid response time (48 ms) and this showed stable durability over 1000 cycles.

III. SENSORS FOR WEARABLE GAIT DETECTION DEVICES

A flexible pressure sensor made of microstructured polydimethylsiloxane film has been demonstrated by Yuan Zhang et al. using a simple colloid self-assembly technique for monitoring human body locomotion [32]. The developed pressure sensor having broad pressure range, 0-4 kPa, exhibited very sensitive detection , fast response time (< 1000 ms) and excellent stability. Flexible strain sensors have been

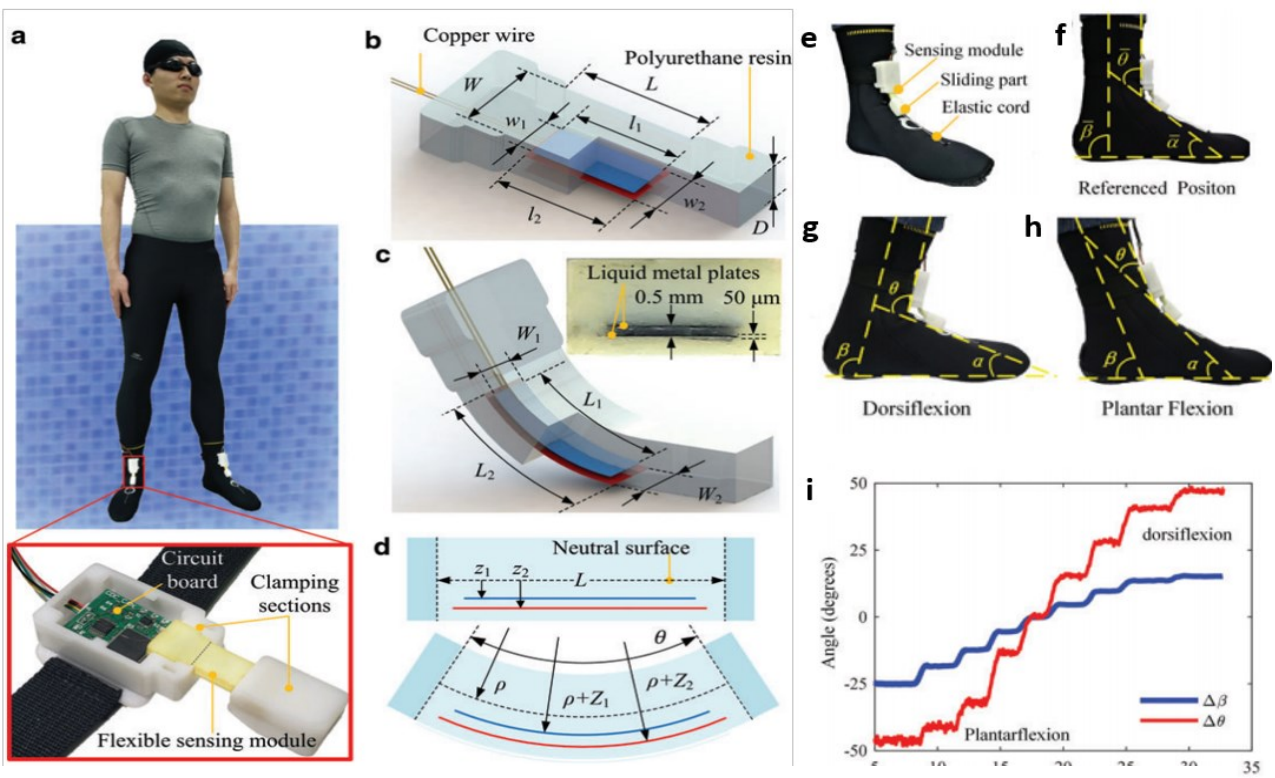

Fig. 4. *(a) Flexible sensor module and its overview. (b) Undeformed and (c) deformed sensor configuration. (d) Schematics of curvature sensor in bend and unbend conditions. (e) installation of sensor, (f) referenced position, (g) dorsiflexion and (h) plantarflexion of joint of the ankle. (i) Measurement of dorsiflexion and plantarflexion [53].*

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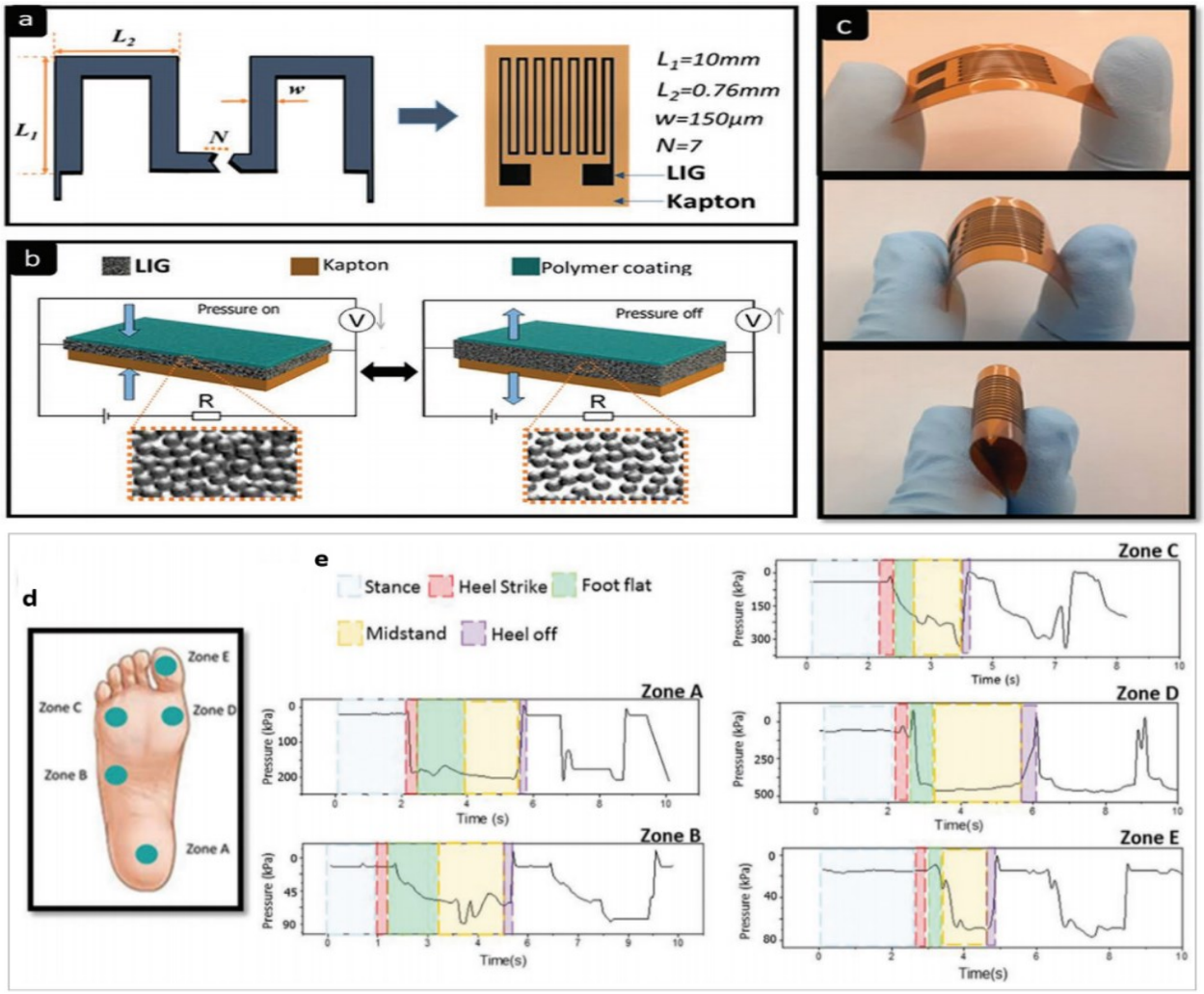

Fig. 5. *Laser-induced graphene (LIG) pressure sensor (a) design and (b) operating principle. (c) Flexible LIG pressure sensor on polyamide (PI) film. (d) Application of LIG pressure sensor on foot at five different positions and (e) recorded sensor signals during a number of gait cycles [54].*
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demonstrated for detecting the deformations of soft tissues [33, 34] and articular joint rotations [35-37]. Recently, flexible electronics based curvature sensors integrated on a flexible wearable substrate have been developed to detect body gestures and joint bending deformations [38]. These strain sensors and curvature sensors are classified as capacitive [39-42] and resistive sensors [36, 21, 46-50]. Several flexible sensor devices have been developed for human locomotion monitoring, however, sensor calibration with the joint rotation angles is not an easy task because of the inconsistent sensor placement location and the nonlinear musculoskeletal deformations [37, 43, 45, 46, 48, 51, 52]. Very recently, Li et al. developed and calibrated a flexible capacitive curvature sensor for monitoring human articular joint movements (Fig. 4) [53]. The sensing principle of this developed sensor is capacitive, i.e., here they relate the bend angle of the sensor with a change in capacitance via deformation curvature. This polyurethane elastomer based device was fabricated via shape deposition molding (SDM) technique. The EGaIn liquid metal was used as capacitive electrode plates.

In a recent work, Kaidarova et al., developed a flexible piezoresistive pressure sensor with a wide pressure measurement range by laser printing technique for gait monitoring (Fig. 5) [54]. This wearable pressure sensor was based on porous graphene conductive electrodes printed on a mechanically flexible PI film is mechanically flexible. In this sensor, pressure stimuli transduce into resistance variations, i.e., the resultant pressure response from the deformed conductive porous graphene arrangement produces resistance variations. In this case, the overall resistance is the sum of the resistance from the piezoresistive material and the graphene electrodes. The developed Laser-Induced Graphene (LIG) sensor satisfies all the functional requirements such as flexibility and reduced weight to implement on foot. As shown in Fig. 5, LIG pressure sensor

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placed on foot recorded measurements in different positions (Zone A (the heel), Zone B (lateral arch), Zone C (first metatarsal head), Zone D (lateral metatarsal), and Zone E (hallux)). The signals were recorded during number of gait cycles (Fig. 5d, e). The amplitude of the recorded signal from the pressure sensor at different points of time (Fig. 5e) shows the highest pressure variation ≈470 kPa (from the Zone D) and least variation ≈84 kPa (from Zone E) when the foot falls OFF and ON on the ground. This sensor provides ~ 20 MPa pressure range. The authors recommend this LIG pressure sensor for other applications including underwater operation, tactile sensor and heart rate monitoring as this sensor can be tailored very easily for a range of applications and monitoring scenarios.

IV. CONCLUSIONS

This article summarises the recent achievements in the development of sensors implemented in wearable devices for physiological parameter and physical activity monitoring that affects WMSDs and gait disorders. Wearable devices are advancing towards high flexibility, cost effectiveness, and better functionality. To realize the non-invasive highly precise real time monitoring of work related and age related health states and disease prevention, researchers have developed numerous wearable sensor systems for gait monitoring, posture and activity recognition, fall risk analysis, plantar pressure measurement and energy expenditure estimation.

The large scale production and application of cost effective wearable devices and sensor systems requires further advances for allowing conformability, increased functionality, and cost effective sensors production techniques (for example, printing techniques). Further research and development activities need to be performed to fully utilize the fundamental physical properties of the biocompatible and natural materials for developing flexible and wearable electronic devices and sensor systems for healthcare monitoring.

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